ADIE Experimental Specification — Digital and Physical Execution Boundaries

Digital and Physical Execution Boundaries

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Status: Draft specification; milestone in progress

Authority: Subordinate to constitutional Sources, approved ADRs and current roadmap

Implementation authorization: None

1. Purpose and decision boundary

Define a bounded, reproducible experiment to test the frozen ADIE reference across one digital and one simulated physical execution boundary:

Security Law Book → Dynamic Authority Derivation → Portable Authority → Independent Enforcement at Every Execution Boundary → Verified Effect → Audit

This specification does not approve architecture, finalize the Security Law Book schema, select a policy engine, authorize production deployment, or claim universal enforcement. Identity/session/permission is not complete dynamic authority. A decision is not execution; an execution receipt is not proof of effect; a log is not proof of complete observation.

2. Objectives and hypotheses

The experiment shall:

- preserve actor, agent, delegation, task, resource, action, audience and scope across handoffs;
- derive deterministic authority from a versioned input/rule snapshot;
- enforce independently at each tested executor boundary;

- reject malformed, mismatched, expired, revoked, stale or replayed authority;
- distinguish decision, admission, execution, observed effect and verification;
- use independent read-back/observation for effect verification;
- reconstruct the operation timeline and expose evidence gaps.

ID	Hypothesis	Falsification
H1	Authority remains bounded across handoff	Boundary accepts altered subject/audience/action/resource/scope
H2	Tested boundary enforces locally	Protected mutation occurs without valid admission
H3	Revocation/expiry follows declared commit rule	Admission after cutoff without approved exception
H4	Execution and effect are separate	Provider success alone becomes **VERIFIED**
H5	Evidence reconstructs the operation	Reviewer cannot correlate request→decision→enforcement→execution→observation
H6	Unknown evidence is not permission/success	Missing/contradictory evidence yields **ALLOW** or **VERIFIED**

3. Experiment scope

Included: contract-level simulation; one controlled digital mutation; one simulated sensor→controller→actuator/process path; authority derivation; portable test artifact; local enforcement; negative/race/bypass/fault scenarios; independent observation; evidence reconstruction.

Excluded: production/customer data; real PLC or safety-critical device; changes to Core or NoredAm/DevTools; policy-engine selection; final authority schema; universal integration coverage; generic emergency override or safety-state behavior.

4. Conceptual topology

```
Untrusted intent → Request normalization → Identity/context references
→ Security Law Book test rules → Dynamic authority derivation
→ ALLOW / DENY / ESCALATE / INDETERMINATE
→ (ALLOW only) portable authority test artifact
├─ Digital boundary validator → Digital executor → independent read-back
└─ Physical boundary validator → simulated controller/actuator
→ independent process observation
→ Effect predicates → Evidence correlation/reconstruction
```

This is a test arrangement, not an approved deployment topology.

5. Stable correlation identifiers

Every run uses unique identifiers: **test_run_id**, **scenario_id/version**, **request_id**, **task_id**, **decision_id**, **authority_artifact_id**, **operation_id**, **execution_id**, **boundary_id**, **resource_id**,

observation_id, **verification_id**, **evidence_record_id**. IDs are correlation keys, not trust proof; issuer and integrity must be recorded.

6. Authority decision test contract

Required inputs:

- accountable principal and authenticated identity reference;
- agent/workload and delegation chain, when applicable;
- tenant/organization/resource ownership context;
- normalized action/resource, executor/audience and task/operation;
- authoritative rule-set ID/version, evaluator version/digest;
- context/risk/runtime facts with provenance and freshness;
- validity and intended effect/postcondition.

Decision is exactly one of: **ALLOW**, **DENY**, **ESCALATE**, **INDETERMINATE**. No implicit allow. **INDETERMINATE** is non-executable for protected operations; **ESCALATE** is not a permit.

Derivation is deterministic for an immutable input/rule/evaluator snapshot. Applicable constraints compose restrictively; explicit prohibition cannot be overridden by a lower-authority grant; missing/conflicting/stale required facts cannot manufacture permission; delegated authority cannot exceed the delegator. Record evaluated constraints, source/version and disposition.

7. Portable authority test artifact

The integrity-protected artifact contains:

- issuer ID, key/version reference and issuer standing;
- principal, agent/workload and delegation chain;
- request/task/operation binding;
- audience/boundary, tenant, resource and action;
- bounded scope and obligations;
- issue/not-before/expiry times and rule/policy version;
- nonce/replay control and revocation reference;
- decision ID, result and reason/constraint references;
- integrity algorithm/key ID and test-run linkage.

It is not self-authorizing. The executor validates integrity, issuer trust, binding, scope, validity, freshness/revocation and local constraints before admission.

8. Boundary enforcement contract

Both boundaries independently:

1. authenticate caller and boundary identity;
2. verify artifact integrity and issuer trust;
3. bind exact request/action/resource/audience/tenant/operation;
4. reject invalid, expired, revoked, replayed or mismatched artifacts;

5. check local capability and domain constraints;
6. record enforcement result/reason;
7. execute only on valid **ALLOW**;
8. distinguish admission from execution;
9. expose operation correlation/state transitions;
10. preserve local domain/safety controls.

Direct/alternate paths are tested. A bypass must be blocked or detected and reported as coverage failure—not counted as enforced.

9. Digital executor profile

Perform one bounded mutation on an isolated test resource. **VERIFIED** requires independent read-back of resource identity, expected version/change, exact value/content digest, no unauthorized collateral mutation and correlation/integrity. Executor response alone is **EXECUTION_REPORTED**, not **VERIFIED**.

10. Simulated physical executor profile

Deterministic simulator fields: sensor value/time/quality; threshold/hysteresis; controller state; actuator command/state; process state (e.g. tank level); independent observation; injected delay/fault.

Example: pump activation is permitted only below a test low-level threshold and with no configured interlock/fault. Thresholds, hysteresis, timing and interlocks are explicit simulator parameters—not ADIE defaults. No physical safety behavior is inferred.

VERIFIED requires fresh/quality-valid sensor evidence, satisfied preconditions/interlocks, valid admission, expected actuator state, process predicate within declared time window, independent correlated observation and no contradictory fault evidence. Command, output and process outcome remain distinct. Failure/unknown yields **FAILED**, **PARTIAL**, **NOT_VERIFIED** or **INDETERMINATE**, never automatic success.

11. Separate state dimensions

Authority: NOT_EVALUATED → ALLOW | DENY | ESCALATE | INDETERMINATE

Execution: NOT_ADMITTED → ADMITTED → QUEUED (optional) → DISPATCHED → RUNNING → COMPLETED | FAILED | PARTIAL | OUTCOME_UNKNOWN

Effect: NOT_STARTED → OBSERVING → VERIFIED | NOT_VERIFIED | CONTRADICTORY | INDETERMINATE

Cancellation: NOT_REQUESTED → REQUESTED → ACKNOWLEDGED | REJECTED | TOO_LATE

Cancellation requested does not prove execution stopped or effect reversed. Retry/compensation is a separately authorized operation with idempotency controls.

12. Revocation and commit ordering

Define one boundary admission/commit point. Compare revocation/cancellation against controlled sequence/commit evidence, not timestamps alone:

- revocation before admission commit: reject;
- revocation after commit: follow explicitly declared in-flight rule and record whether execution continues/stops/becomes unknown;
- expiry before commit: reject;
- cancellation acknowledgement does not prove physical reversal;
- late revocation cannot be represented as retroactively preventing committed effect.

Report the chosen rule and observed behavior; do not generalize to every subsystem.

13. Evidence record

Each scenario records:

- scenario/run/configuration; rule source/version and evaluator digest;
- context facts/provenance references;
- actor, agent, delegation, tenant, resource, action, audience;
- request/task/decision/artifact/operation/execution IDs;
- decision, diagnostics and evaluated constraints;
- boundary enforcement result; event sequence and commit ordering;
- executor response/raw state transitions;
- observation source, freshness, quality and result;
- postcondition and verification result;
- integrity references, missing/contradictory events and coverage gaps;
- status **PASS**, **FAIL**, **BLOCKED**, **NOT_RUN**, reviewer/approval references.

Minimize sensitive raw data and follow approved privacy/retention rules.

14. Scenario suite

ID	Scenario	Expected result
D01	Valid digital mutation	Admission + exact independent read-back
D02	Wrong resource/action/audience	Reject; no mutation
D03	Invalid integrity/issuer	Reject; no mutation
D04	Expired/not-yet-valid	Reject
D05	Revoked before commit	Reject
D06	Replay/duplicate	Prevent or detect duplicate side effect
D07	Delegation expansion	Reject; child cannot exceed parent

ID	Scenario	Expected result
D08	Missing/conflicting context	Indeterminate/deny; no protected execution
D09	Success response but mismatched read-back	Not verified/contradictory
D10	Direct bypass	Block or report boundary coverage failure
P01	Valid simulated pump	Actuator and process predicates independently verified
P02	Stale/invalid sensor	Reject/indeterminate; no unsafe command
P03	Interlock/fault active	Local rejection; no effect claimed
P04	Authority mismatch/revocation	Reject before commit
P05	Command accepted, actuator does not move	Execution reported; effect not verified
P06	Actuator moves, process predicate fails	Partial/not verified
P07	Observer unavailable	Indeterminate
P08	Contradictory observations	Contradictory; simulator disposition applies
X01	Missing audit event	Gap detected; completeness assertion fails
X02	Clock skew/out-of-order	Sequence/commit reconstructs or flags ambiguity
X03	Revocation/cancellation race	Declared commit semantics observed
X04	Evidence tampering	Integrity verification fails
X05	Partial execution/timeout	Partial/outcome unknown; no false success
X06	Cross-boundary correlation	Actor/constraints/operation linkage retained

These identifiers supplement—not replace—Security Law Book T01–T20 and expanded S01–S30.

15. Test oracle and status

Compare independently: authority decision; boundary admission; execution; observed effect; verification; evidence integrity/completeness.

- **PASS**: all assertions and evidence requirements pass.
- **FAIL**: assertion violated or unauthorized effect occurs.
- **BLOCKED**: precondition/environment prevents meaningful run.
- **NOT_RUN**: not executed.
- **INDETERMINATE**: evidence/outcome cannot support conclusion; never a pass.

No aggregate score may hide unauthorized effect, bypass, false verification or evidence-integrity failure.

16. Metrics

Report boundary paths identified/tested; blocked/detected bypasses; unauthorized effects (target zero); invalid/revoked/replayed artifacts accepted (target zero); false **VERIFIED** (target zero); effect-verification coverage/independence; revocation ordering; evidence completeness/reconstruction; indeterminate outcomes; stage latency; simulator fault coverage. These are experiment criteria, not production SLAs/certification.

17. Exit criteria

Ready for review only when:

- both boundaries/trust boundaries documented;
- scenario inputs, outcomes and oracles deterministic;
- decision/enforcement/execution/effect states separate;
- independent digital read-back and physical observation defined;
- negative, race, bypass and evidence-fault cases included;
- reconstruction exposes all evidence gaps;
- unexecuted cases and limitations explicit;
- no production or architecture adoption claim.

18. Limitations

Simulation is not physical safety/field reliability evidence. Test adapters cannot establish universal coverage. Observation independence may be imperfect. Revocation semantics are boundary/commit-specific. Policy lifecycle, authority hierarchy, key lifecycle and canonical artifact remain unresolved. Post-transfer sovereignty is out of scope. No runtime tests are claimed by this specification.

19. Deliverables when implementation is separately authorized

1. Versioned topology/configuration.
2. Deterministic digital and physical simulator.
3. Authority fixtures/rule set.
4. Boundary validators/adapters.
5. Scenario runner/test oracle.
6. Evidence records/integrity checks.
7. Reconstruction report and gap register.
8. Verification summary and approval record.

20. Next gate

This specification is subordinate to constitutional Sources, approved ADRs and roadmap. Any authority conflict blocks progression.

Next gate: explicit authorization for bounded experiment implementation and test environment. Then:

Specification approval → Implementation → Compilation (where applicable) → Verification → Approval → Next milestone

No production deployment, architecture freeze or broad subsystem integration follows automatically.

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